In [ ]:
```
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```

In [2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [4]:
```
def sigmoid(x):
    return 1 / (1 + np.exp(-x))
```

In [5]:
```
def relu(x):
    return np.maximum(0, x)
```

In [6]:
```
def tanh(x):
    return np.tanh(x)
```

In [7]:
```
def softmax(x):
    return np.exp(x) / np.sum(np.exp(x))
```

In [8]:
```
# Create x values
x = np.linspace(-10, 10, 100)
```

In [14]:
```
# Add common axis labels and titles
fig.suptitle('Common Activation Functions')
for ax in axs.flat:
    ax.set(xlabel='x', ylabel='y')
```

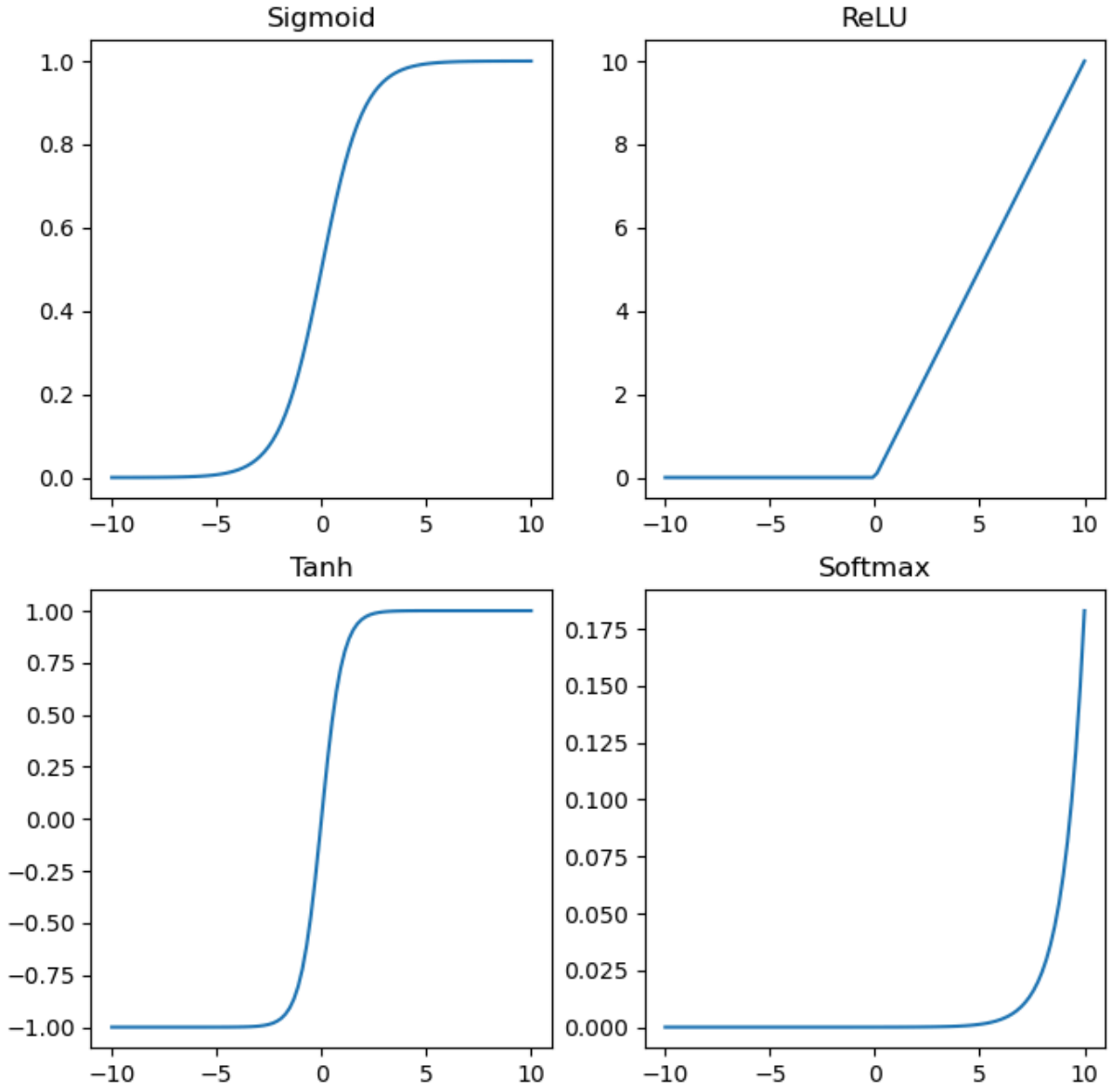
In [12]:
```
# Adjust spacing between subplots
plt.subplots_adjust(left=0.1, bottom=0.1, right=0.9, top=0.9, wspace=0.4, hspace=0.
```

```
<Figure size 640x480 with 0 Axes>
```

In [13]:
```
# Show the plot
plt.show()
```

In [9]:
```
# Create plots for each activation function
fig, axs = plt.subplots(2, 2, figsize=(8, 8))
axs[0, 0].plot(x, sigmoid(x))
axs[0, 0].set_title('Sigmoid')
axs[0, 1].plot(x, relu(x))
axs[0, 1].set_title('ReLU')
axs[1, 0].plot(x, tanh(x))
axs[1, 0].set_title('Tanh')
axs[1, 1].plot(x, softmax(x))
axs[1, 1].set_title('Softmax')
```

Out[9]:
```
Text(0.5, 1.0, 'Softmax')
```



In [ ]: